



# AI Lifecycle Security & Human Oversight

Chapter 05 · Episode 01 · Domain 1.3 — The MDLC End-to-End

---



# Learning Objectives

1. Map all seven MDLC phases to their primary controls
2. Distinguish HITL, HOTL, and governance oversight by risk
3. Apply lifecycle risk mapping to exam scenarios

# Phases 1–2: Use Case & Data

## PHASE 1 · USE CASE

Risk-appetite definition

Impact assessment (required)

Regulatory scoping

No documented purpose = unmanaged risk

## PHASE 2 · DATA

Classification & minimization

Integrity & verification

Provenance

Model security is decided here

# Phases 3–4: Development & TEVV

## PHASE 3 · DEVELOPMENT

Threat-model the architecture

OWASP LLM Top 10 · MITRE ATLAS

Secure training infra

Version control + checkpoint signing

## PHASE 4 · TEVV ★

Accuracy + bias testing

Adversarial robustness

Hallucination testing

Fail TEVV → no deployment

# Phases 5–7: Deploy, Monitor, Iterate

## PHASE 5 · DEPLOYMENT

RBAC + encryption

Prompt guardrails

Canary rollout + auto-rollback

The attack surface is now live

## PHASES 6–7 · RUN

Drift & log monitoring

Cost/abuse signals + IR

Human review loops

Retrain → re-run TEVV

# Three Human-Oversight Models



## HITL

HIGH risk

Human approves before action

Identity, finance, discipline



## HOTL

MEDIUM risk

AI acts, human can intervene

Fraud, alert triage



## GOVERNANCE

LOW risk

Humans set policy only

Log/spam classification

1 Use Case

PASS

2 Data

WARN

3 Dev

PASS

4 TEVV

FAIL

5 Deploy

PASS

6 Monitor

WARN

7 Iterate

PASS

## HOW TO RUN

### WINDOWS

```
python lifecycle_risk_scanner.py
```

### MACOS/LINUX

```
python3 lifecycle_risk_scanner.py
```

**TRY IT** Set `PHASE_CONFIG[4]['adversarial_score'] = 85` and observe the TEVV gate passing and the pipeline completing.

`lifecycle_risk_scanner.py`

pure Python 3.10+, no external libraries

# Demo: Lifecycle Risk Scanner

Scans a model config against all seven MDLC phases — a Phase-4 TEVV failure blocks the pipeline

# Lifecycle Risk Mapping (Scenario → Phase)



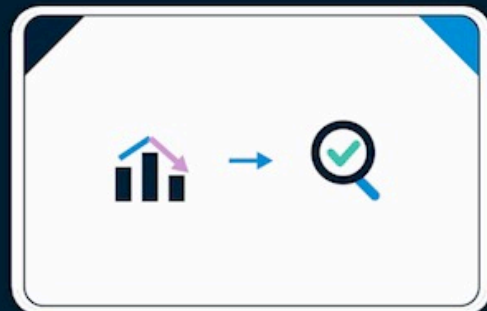
## **BLIND SPOTS**

→ Phase 2 data balancing



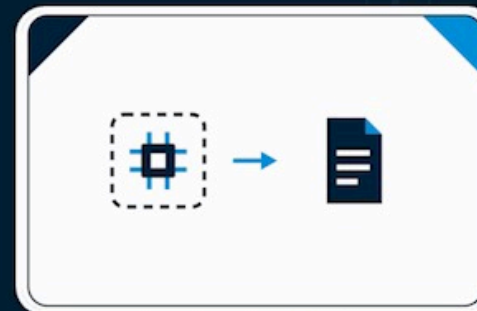
## **UNDISCOVERED BIAS**

→ Phase 4 TEVV



## **DRIFT AFTER CHANGE**

→ Phase 6 monitoring



## **SHADOW DEPLOY**

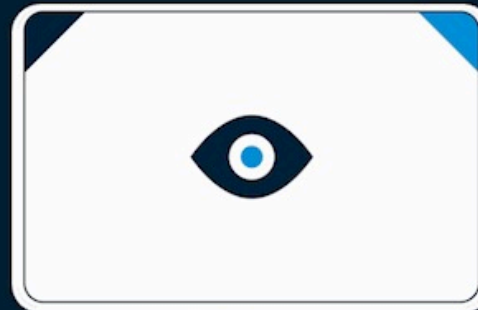
→ Phase 1 use case

# Exam Tips



## **MDLC**

Organizes all of Domain 1



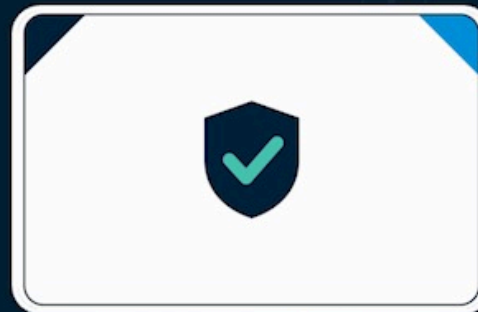
## **OVERSIGHT**

HITL/HOTL/gov by risk



## **TEVV**

A gate, not a checkbox



## **SECURITY**

At every phase